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Symbolic Wavefunctions and Entropic Fractals: A Structural–Analogical Heuristic for Socio–Symbolic Dynamics

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Abstract

We advance a formal–analogical framework that conceives social organization as an *entropically driven fractal dynamics*. A symbolic Schrödinger–like representation uses a normative potential $V(\mathbf{x})$ for cultural constraints and a symbolic mass m_s for identity inertia; the symbolic wavefunction $\psi_s(\mathbf{x}, t)$ induces densities $|\psi_s|^2$ on a socio–symbolic space. Crucially, the quantum machinery is treated as a *structural analogy*—not a physical mechanism. We articulate falsifiable signatures (scaling exponents (West et al. 1997; Bettencourt et al. 2007), cascade statistics (Bak et al. 1987; Clauset et al. 2009; Stumpf and Porter 2012; Alstott et al. 2014), order–effects (Busemeyer and Bruza 2012; Haven and Khrennikov 2013)), outline estimation of $V(\mathbf{x})$ and m_s , and present a concise SWOW–style proof–of–concept. The contribution is a disciplined mapping that preserves explanatory payoff for social emergence and cognitive–affective coupling while avoiding category mistakes (Hesse 1966; Cartwright 1983; Frigg and Hartmann 2020).

Keywords: philosophy of science; structural analogy; complex systems; fractals; entropy; quantum-like models; socio-symbolic dynamics; methodology

1 Introduction

We propose a structural–analogical formalism to model socio–symbolic dynamics via a symbolic wavefunction $\psi_s(x, t)$ evolving under a normative potential $V(x)$ and a symbolic mass m_s . The approach converses with small–world (Watts and Strogatz 1998)

QM element	Social counterpart
$\psi(x, t)$	$\psi_s(x, t)$: socio-symbolic amplitude
$ \psi ^2$	$ \psi_s ^2$: density of manifestation
$-\frac{\hbar^2}{2m}\nabla^2 + V(x)$	$-\frac{\hbar^2}{2m_s}\nabla^2 + V(x)$
∇^2	diffusion/transition propensity in symbolic space
m	m_s : identity inertia/rigidity
$V(x)$	normative-cultural field/cost of deviation
Interference	non-additive discursive coupling/order effects
Tunnelling	improbable transitions across norms

Table 1 Structural mapping from quantum formalism to socio-symbolic dynamics.

and scale-free organization (Barabási and Albert 1999), self-organized criticality (Bak et al. 1987), and scaling regularities in complex systems (West et al. 1997; Bettencourt et al. 2007), while aligning with philosophical accounts of models and analogies (Hesse 1966; Cartwright 1983; Frigg and Hartmann 2020). Our goal is to maintain falsifiability and operational clarity without physical reduction.

2 Method and Epistemic Status

We take the symbolic evolution

$$i\hbar\partial_t\psi_s = \left[-\frac{\hbar^2}{2m_s}\nabla^2 + V(x)\right]\psi_s, \quad x \in X \subset \mathbb{R}^d, \quad (1)$$

as an *analogy of structure* (Hesse 1966). Here, $V(x)$ encodes normative-cultural constraints; m_s encodes identity inertia; $|\psi_s|^2$ is a density of manifestation. Interference patterns correspond to non-additivity/order effects (Busemeyer and Bruza 2012; Haven and Khrennikov 2013); tunnelling analogs capture barrier crossing under strong constraints. No ontological commitment to quantum substrates is made.

2.1 Mapping (QM $\rightarrow$ Social)

3 Testable Signatures

Variable	Signature	Metric/Operationalization
m_s	slower trajectories; rigidity	path entropy; embedding drift; Hurst exponent
$V(x)$	alignment; confinement; tunnelling	GP/regression for V ; change-points; transition hazard
$ \psi_s ^2$	dense pockets; non-additivity	cluster overlap; Jensen–Shannon divergence; order to chaos
∇^2	smoothing/propagation	Laplacian; total variation; diffusion coefficients
SOC cascades	heavy-tailed sizes	power-law inference (Clauset et al. 2009; Stumpf and Porter 2012)
Metastability	dwelt + jumps	dwelt-time; CUSUM/PELT
Interference	order-dependence	quantum-like decision paradigms (Busemeyer and Bruza 2004)

4 Estimation Blueprint

Construct X using language embeddings/behavioral features; estimate $V(x)$ with penalized regression or Gaussian processes; infer m_s from identity persistence and dwelt times; validate with diffusion baselines and synthetic recovery.

5 Proof-of-Concept (SWOW-style)

We illustrate rank–frequency structure and salient cue $\rightarrow$ response links using a small SWOW-style subset, as in large-scale word association studies (Watts and Strogatz 1998; Barabási and Albert 1999).

6 Objections and Limits

No physical reduction is claimed; the analogy is structural (Hesse 1966; Cartwright 1983). Identifiability of $V(x)$ and m_s requires longitudinal data; ethical notes include transparency on assistive AI and dataset licenses.

7 Conclusion

A disciplined mapping from quantum formalism to socio-symbolic dynamics with falsifiable signatures (Clauset et al. 2009; Stumpf and Porter 2012; Alstott et al. 2014) and a practicable estimation path from data to explanation.

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Data Availability. Supplementary materials and figure-generation code are provided with the submission; SWOW data are subject to original licensing.

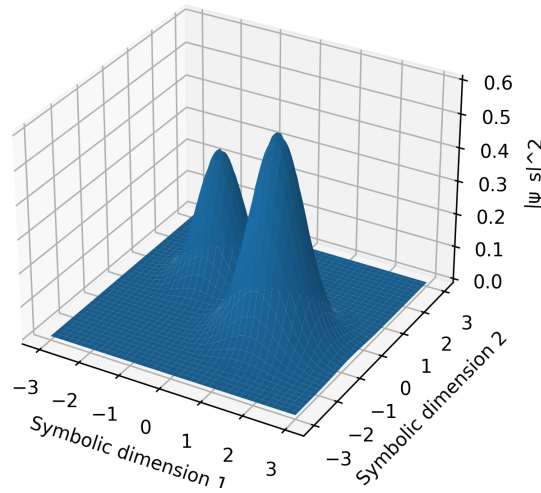


Fig. 1 Symbolic wavefunction density $|\psi_s|^2$ over a socio-symbolic space (3D surface; illustrative).

Competing Interests. The author declares no competing interests.

References

- Alstott J, Bullmore E, Plenz D (2014) Powerlaw: A Python package for analysis of heavy-tailed distributions. PLoS ONE 9(1):e85777. <https://doi.org/10.1371/journal.pone.0085777>
- Bak P, Tang C, Wiesenfeld K (1987) Self-organized criticality: An explanation of $1/f$ noise. Physical Review Letters 59(4):381–384. <https://doi.org/10.1103/PhysRevLett.59.381>
- Barabási AL, Albert R (1999) Emergence of scaling in random networks. Science 286(5439):509–512. <https://doi.org/10.1126/science.286.5439.509>
- Bettencourt LMA, Lobo J, Helbing D, et al (2007) Growth, innovation, scaling, and the pace of life in cities. Proceedings of the National Academy of Sciences 104(17):7301–7306. <https://doi.org/10.1073/pnas.0610172104>
- Bussemeyer JR, Bruza PD (2012) Quantum Models of Cognition and Decision. Cambridge University Press, Cambridge, <https://doi.org/10.1017/CBO9780511997716>
- Cartwright N (1983) How the Laws of Physics Lie. Clarendon Press, Oxford
- Clauset A, Shalizi CR, Newman MEJ (2009) Power-law distributions in empirical data. SIAM Review 51(4):661–703. <https://doi.org/10.1137/070710111>

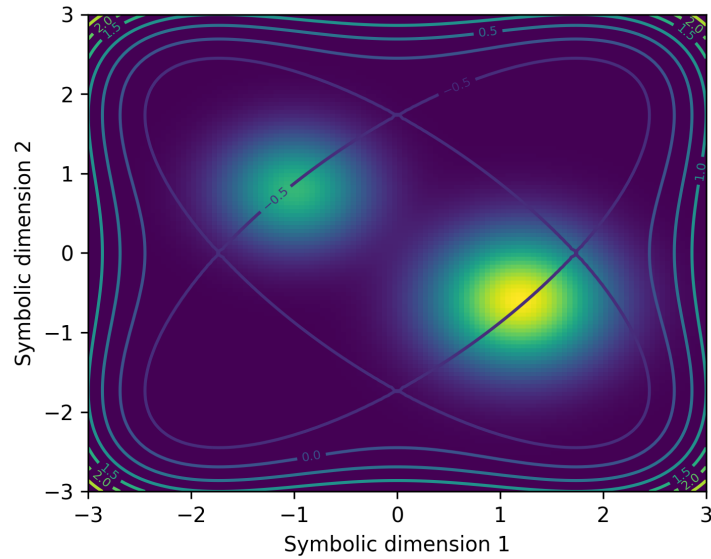


Fig. 2 Density map $|\psi_s|^2$ with normative potential contours $V(x)$ (illustrative).

Frigg R, Hartmann S (2020) Models in science. In E. N. Zalta (ed.), *The Stanford Encyclopedia of Philosophy*, Fall 2020 Edition, URL <https://plato.stanford.edu/archives/fall2020/entries/models-science/>, metaphysics Research Lab, Stanford University

Haven E, Khrennikov A (2013) *Quantum Social Science*. Cambridge University Press, Cambridge, <https://doi.org/10.1017/CBO9781139003552>

Hesse M (1966) *Models and Analogies in Science*. University of Notre Dame Press, Notre Dame, IN

Stumpf MPH, Porter MA (2012) Critical truths about power laws. *Science* 335(6069):665–666. <https://doi.org/10.1126/science.1216142>

Watts DJ, Strogatz SH (1998) Collective dynamics of ‘Small-World’ networks. *Nature* 393:440–442. <https://doi.org/10.1038/30918>

West GB, Brown JH, Enquist BJ (1997) A general model for the origin of allometric scaling laws in biology. *Science* 276(5309):122–126. <https://doi.org/10.1126/science.276.5309.122>

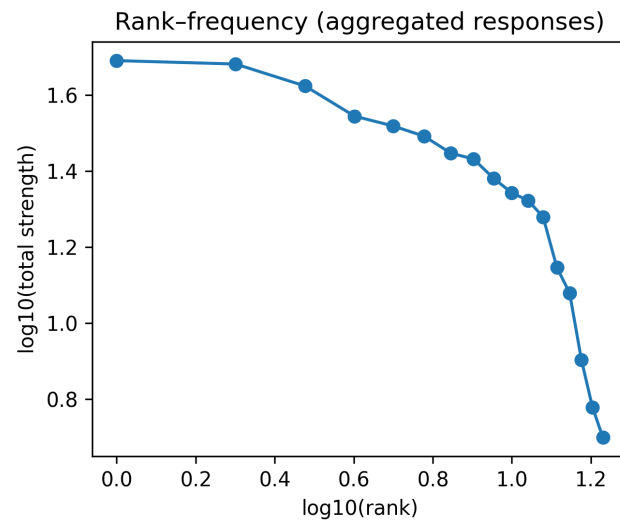


Fig. 3 Rank-frequency (log-log) for aggregated responses (SWOW-style subset).